

FUNDAMENTALS OF COMPUTER



What is Computer?

A computer is an electronic machine that takes an input, processes it to produce the desired output. Every computer is a combination of hardware and software. The physical components of a computer that can be seen and touched form the hardware. For example, CPU, monitor, keyboard, printer etc., are hardware or peripheral devices. The input to a computer is given in the form of instructions. These set of instructions that we give to the computer to perform a particular task constitutes a program. Many such programs together form a software for the computer. Operating system, Antivirus, MS Office, Computer games are all software applications.

The word “Computer” is derived from the Latin word “Computare” which literally means “to calculate” or “to sum up”. Historically, computers were primarily used for complex calculations, though their role has expanded significantly today.

Desktop Computer Components



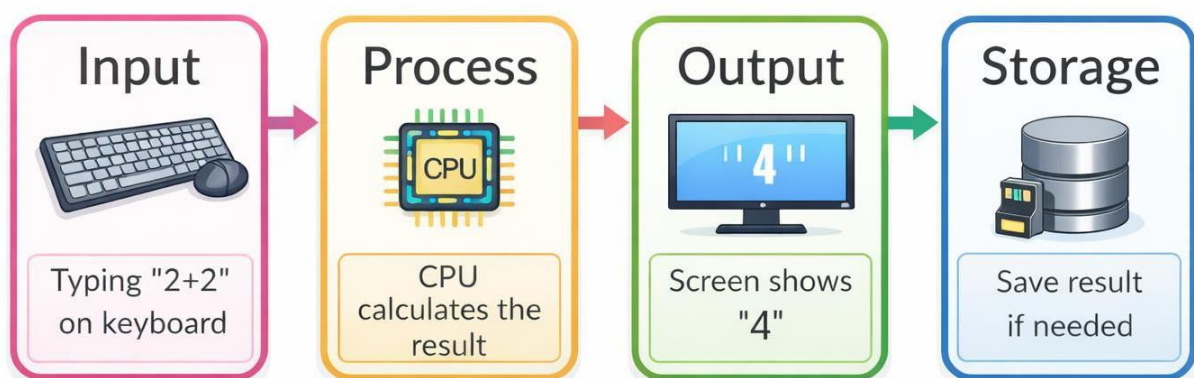
The computer follows a basic cycle of operation

Input: Data and instructions are received through input devices (e.g. keyboard, mouse.)

Processing: The raw data is transformed into meaningful information using programmed instructions.

Output: The results are displayed or presented through output devices (e.g. monitor, printer.)

Storage: Information can be saved for future reference.



INPUT AND OUTPUT DEVICES

INPUT DEVICES:

An input device is used to get data or instructions from the user. This data is then passed on to CPU for processing so as to produce the desired result. Although keyboard and mouse are the two common input devices, other devices such as Optical Character Recognition (OCR), Magnetic Ink Character Recognition (MICR), and mark sense reader, etc., are also used as per our requirement.

BASIC COMPUTER INPUT DEVICES



KEYBOARD



MOUSE



HEADSET (WITH MIC)



WEBCAM

OUTPUT DEVICES:

The result of Input processing is Known as Output. Output device is hardware component that display or produce the result of processed data. Output devices receive information from the CPU and present it to the user in the desired form. Some of the output devices are monitor, printers, plotters, etc.

BASIC COMPUTER OUTPUT DEVICES



MONITOR



PRINTER



PROJECTOR



COMPUTER SPEAKERS

CPU– CENTRAL PROCESSING UNIT

What is Processing?

Processing is the task of converting input data into meaningful output using instructions. This work is done by the CPU — **the brain of the computer**.

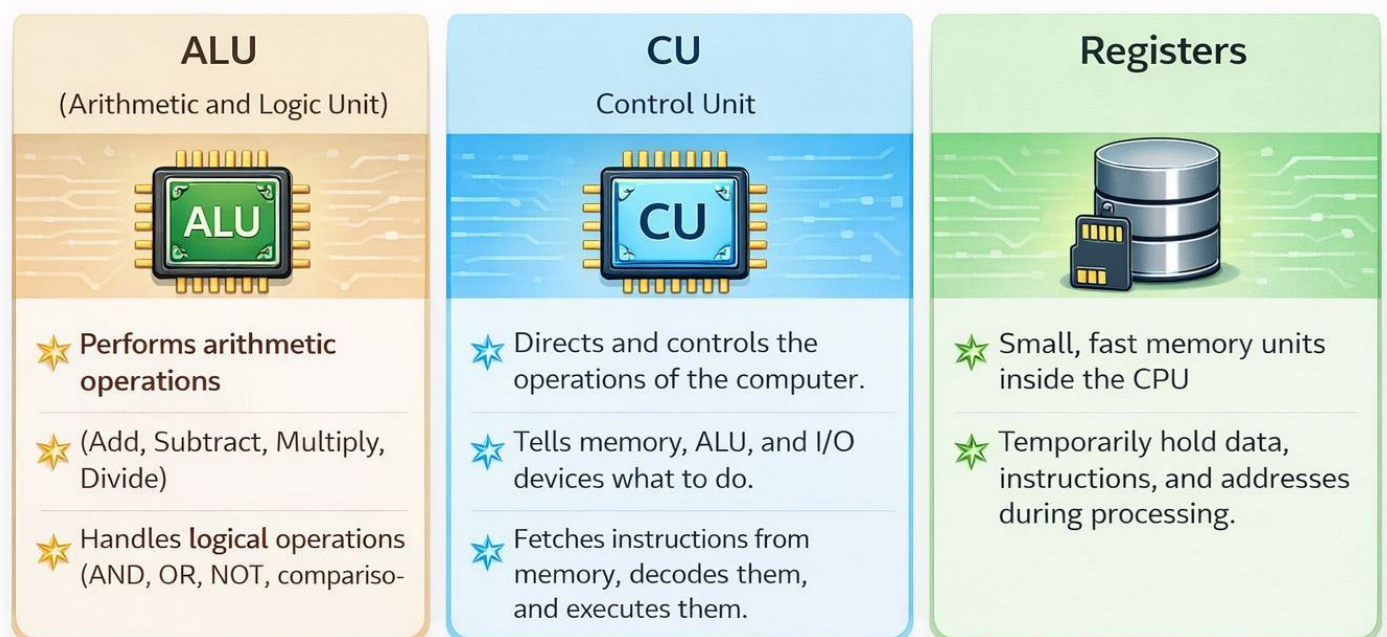
What is CPU?

CPU (Central Processing Unit) is the main component in a computer that performs all the calculations and logical operations. It processes instructions, manages data, and controls the flow of information.



As mentioned earlier, CPU is the 'Brain of your computer'. This is because it processes or executes the instructions given to the computer. Any type of instruction given to the computer using any of the input devices has to be sent to the CPU for execution. In a microcomputer, the CPU is based on a single chip called the microprocessor.

Main Components of CPU:



MEMORY UNIT

Memory is an essential part of a computer system. It is used to store data, instructions, and results temporarily or permanently so that the computer can perform different tasks efficiently.

A computer uses memory to:

- Store input data
- Store instructions (programs)
- Store output results



What is Storage?

Storage refers to the part of the computer system that stores data and instructions for current or future use. It allows users to save, retrieve, and manage digital information whenever required.

Storage is required because a computer needs a place to keep data even when it is not being processed.

Why is Storage Important?

Storage plays a very important role in computer systems because it:

- Stores files, documents, images, videos, and software
- Helps in saving data for future use
- Retains data even when the computer is switched off (non-volatile storage)
- Supports both temporary working data and permanent data storage

Types of Storage:

1. Primary Storage (Main Memory):

Primary memory is the memory that is **directly accessed by the Central Processing Unit (CPU)**. It is also known as **main memory** or **internal memory**.

Primary memory stores the data and instructions that are currently being used by the computer. When the computer is switched off, most of the data stored in primary memory is lost.

Characteristics of Primary Memory

- Very fast in speed
- Directly connected to the CPU
- Limited storage capacity
- More expensive than secondary memory
- Mostly volatile in nature

Primary memory is of **two types**:

- **RAM (Random Access Memory)**: Stores data that is currently in use.
- **ROM (Read-Only Memory)**: Stores permanent instructions like boot process.



2. Secondary Storage (Permanent Memory):

Primary memory is very fast but **expensive** and has **limited capacity**. Therefore, computers require additional memory to store large amounts of data. This additional memory is called **secondary memory** or **auxiliary storage**.

Secondary memory is not directly accessed by the CPU and is used for **long-term storage** of data and programs.

Characteristics of Secondary Memory

- Large storage capacity
- Low cost compared to primary memory
- Slower than primary memory
- Non-volatile in nature
- Data is not lost when power is switched off
- Examples of Secondary Memory Devices

Examples:



Measuring Memory

Unit	Size
1 Bit	Smallest unit (0 or 1)
1 Byte	8 Bits
1 KB (Kilobyte)	1024 Bytes
1 MB (Megabyte)	1024 KB
1 GB (Gigabyte)	1024 MB
1 TB (Terabyte)	1024 GB

COMPONENTS OF A COMPUTER

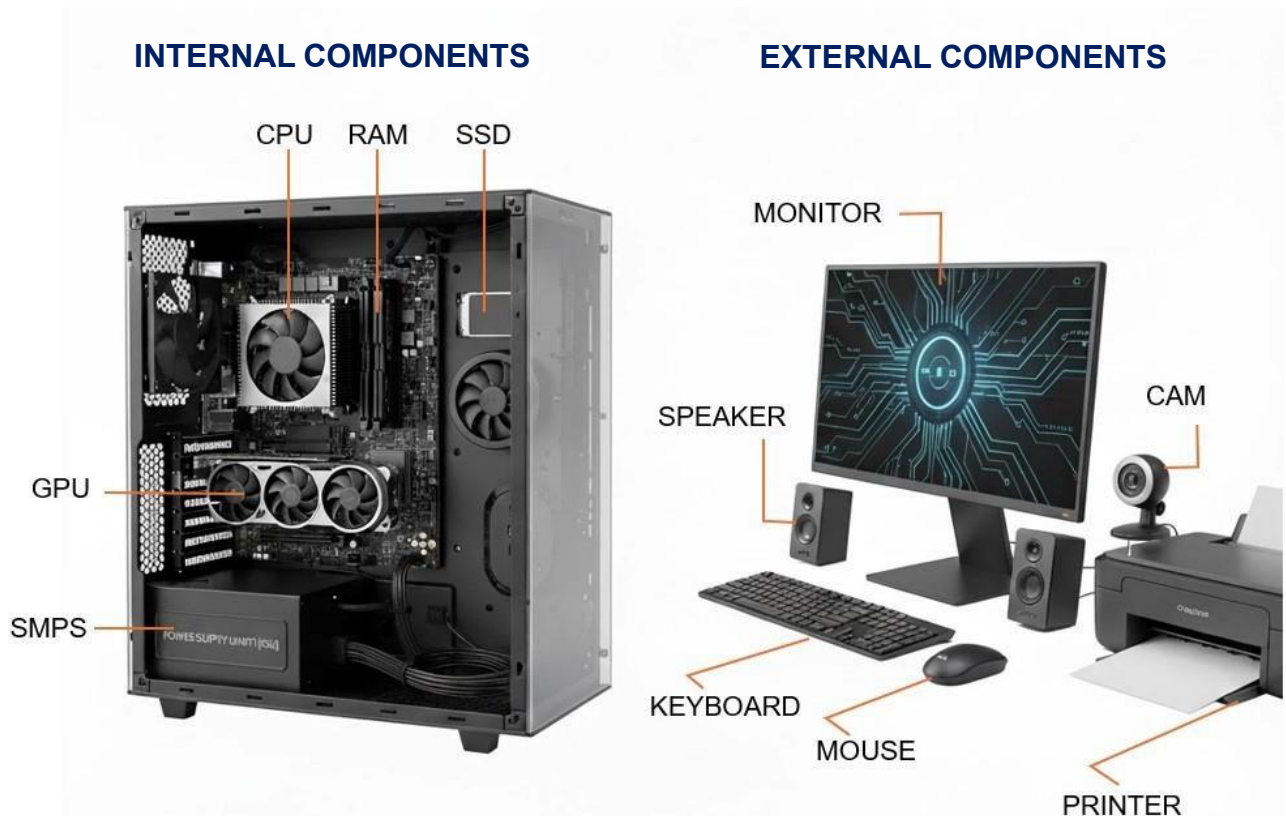
A Computer has two Key Components: Hardware and Software

What is Hardware?

Computer Hardware refers to the **physical components** of a computer system — the parts you can **touch** and **see**.

These components form the basic structure of a computer and are responsible for performing all the operations such as input, processing, storage, and output of data.

Without hardware, a computer system cannot function, because hardware provides the physical platform on which software runs. Hardware and software work together, hardware performs tasks according to the instructions given by software.



What is Software?

Software is a **set of instructions, programs, or data** used to operate computers and perform specific tasks. Unlike hardware, **software is not physical** — you **cannot touch** it.

Software is what tells the hardware what to do. Software controls the working of computer hardware and enables users to perform different tasks such as typing documents, browsing the internet, calculations, designing, and communication.

SYSTEM SOFTWARE



OPERATING SYSTEMS (e.g., Windows, Linux)

-  DEVICE DRIVERS
- FIRMWARE 
- UTILITIES

APPLICATION SOFTWARE



PRODUCTIVITY SUITES

- WEB BROWSERS
- GAMES
- MOBILE APPS
- GAMES
- GRAPHICS & VIDEO EDITORS
- MOBILE APPS

System Software

System software includes general programs written for a computer. It consists of pre-written programs and documentation supplied by the manufacturer along with the computer. These programs are held permanently in the machine. The primary objectives of this software are to:

- (i) enhance the efficiency of hardware utilization, and
- (ii) make computers simple to use.

Operating System

An operating system is the most important system software and it is an essential software to operate the computer system. An operating

Operating system like Microsoft Windows, Linux are examples of system software.

system manages computer's resources very effectively, takes care of scheduling multiple jobs for execution and manages the flow of data and instructions between the input/output units and the main memory.

The first operating system, called batch processing operating system was developed for the second-generation computers. This operating system executes jobs serially one after another from a batch of jobs submitted for execution. The central processing unit is kept busy only during the processing cycle of a job and remains idle during the input and output operations. A multi programming operating system handles multiple jobs simultaneously by overlapping the input/ output and processing cycles of various jobs.



Application Software

Application software is written to perform a specific task or process, such as accounting, payroll, mailing list, result preparation and so on. The majority of application software is written in high-level languages.

Assuming that the task to be carried out has been correctly identified, carefully defined, the prospective user will come across the following alternative sources for this application software.



TYPES OF COMPUTERS

Computers are amazing electronic devices that help us learn, play, work, and connect with the world. They store information, process data, and produce results based on our instructions. With artificial intelligence (AI), some computers can even learn from data! Computers come in different types based on their size and data-handling capabilities. This guide explains these types in a way that's easy to understand, with examples relevant to students.

Types of Computers by Size

Computers vary in size, from huge machines to pocket-sized devices. Here are the main types:

TYPE OF COMPUTER BY SIZE



1. Supercomputers

Supercomputers are the speed demons of the computer world, performing billions of calculations per second. They're like the Hulk of tech, tackling huge tasks like predicting weather or designing spacecraft. **Features:**

- Lightning-fast, thanks to thousands of processors teaming up.

- Super expensive and massive, often taking up entire rooms.
- Used for big-deal research, like studying stars or curing diseases.

Examples: Frontier (helps with climate research), Fugaku (aids medical breakthroughs).

For Students: Supercomputers power the weather apps you check to see if you need an umbrella for recess.

Fun Fact: The world's fastest supercomputer, Frontier, can do a quintillion calculations per second—that's a 1 with 18 zeros!

(1000000000000000000/s)

2. Mainframe Computers

Mainframes are big machines that juggle tons of tasks and users at once. Think of them as a school librarian helping hundreds of kids find books at the same time.

- **Features:**
 - Handles thousands of users and programs simultaneously.
 - Huge storage and speedy processing for massive data.
 - Used by banks, airlines, and big companies to manage records.
- **Examples:** IBM zSeries (keeps bank transactions secure).
- **For Students:** When you buy a game online, mainframes make sure your payment goes through safely.
- **Fun Fact:** Mainframes can run for years without shutting down, like a teacher who never takes a day off!

3. Minicomputers

Minicomputers are mid-sized machines that support multiple users (4–200) for tasks like accounting or billing. They're smaller than mainframes but bigger than your laptop.

- **Features:**
 - More portable than mainframes but still hefty.
 - Reliable for medium-sized jobs.
 - Used in schools, businesses, or labs for specific tasks.

- **Examples:** PDP-11 (an older model used in universities).
- **For Students:** Minicomputers might track your school's attendance or grades.
- **Fun Fact:** Minicomputers were nicknamed "minis" in the 1960s, like the mini skirts of that era!

4. Workstation Computers

Workstations are high-powered computers for one user, built for heavy-duty tasks like creating animations or designing buildings. They're like a turbo-charged art studio for tech work.

- **Features:**
 - Fast processors, tons of memory, and awesome graphics.
 - Perfect for video editing, 3D modeling, or simulations.
 - Stronger than a regular PC but smaller than a mainframe.
- **Examples:** HP Z Workstations (used by engineers or animators).
- **For Students:** Workstations create the special effects in your favorite movies or video games.
- **Fun Fact:** Pixar uses workstations to render animated films like *Toy Story*, frame by frame!

5. Personal Computers (PCs)

PCs are built for one person to use at home, school, or work. This includes desktops, laptops, tablets, and smartphones—stuff you probably use all the time.

- **Features:**
 - Small, affordable, and easy to use.
 - Great for homework, gaming, browsing, or video calls.
 - Comes with a CPU, memory, and input/output devices like keyboards and screens.
- **Examples:** Dell laptops, Apple iMac, iPads, Samsung Galaxy phones.

- **For Students:** You use PCs for writing essays, playing Minecraft, or watching YouTube tutorials.
- **Fun Fact:** The first “personal computer,” the Altair 8800, came as a kit you had to build yourself in 1975!

6. Server Computers

Servers store and share data or programs over a network. They’re like a school cafeteria serving up data to lots of devices at once.

- **Features:**
 - Run 24/7 to deliver services like websites or email.
 - Built to share resources with many client computers.
 - Manage data for multiple devices or users.
- **Examples:** Web servers (host sites like Google), database servers (store school records).
- **For Students:** When you pull up a website for a school project, servers send that webpage to your device.
- **Fun Fact:** Servers “talk” to each other so fast that a website can load in milliseconds, quicker than you can blink!
- Supercomputers
- Mainframe Computers
- Minicomputers
- Microcomputers (PCs)

Types of Computers based on data-handling capabilities



1) Analog Computer

Analog Computers are particularly designed to process analog data. Continuous data that changes continuously and cannot have discrete values are called analog data. So, an analog computer is used where we don't need exact values or need approximate values such as speed, temperature, pressure, etc. It can directly accept the data from the measuring device without first converting it into numbers and codes. It measures the continuous changes in physical quantity. It gives output as a reading on a dial or scale. For example, speedometer, mercury thermometer, etc.

Characteristics:

- Handles **continuous data**.
- **Real-time data processing** without needing to convert it into digital form.
- Used in applications where **approximate values** are sufficient.

Examples:

- **Speedometer** (for measuring vehicle speed)
- **Thermometers** (mercury thermometer)

2) Digital Computer

Digital computers are designed in such a way that they can easily perform calculations and logical operations at high speed. It takes raw data as input and processes it with programs stored in its memory to produce the final output. It only understands the binary input 0 and 1, so the raw input data is converted to 0 and 1 by the computer and then it is processed by the computer to produce the result or final output. All modern computers, like laptops, desktops including smartphones are digital computers.

Characteristics:

- Processes data in the form of **binary digits** (0s and 1s).
- Handles **precise, accurate data** and complex operations, including calculations, logical operations, and storage.
- **Fast data processing** and capable of running various applications from entertainment to business management.

Examples:

- **Laptops**
- **Desktops**
- **Smartphones**

3) Hybrid Computer

As the name suggests hybrid, which means made by combining two different things. Similarly, the hybrid computer is a combination of both analog and digital computers. Hybrid computers are fast like analog computers and have memory and accuracy like digital computers. So, it has the ability to process both continuous and discrete data. For working when it accepts analog signals as input then it converts them into digital form before processing the input data. So, it is widely used in specialized applications where both analog and digital data are required to be processed. A processor which is used in petrol pumps that converts the measurements of fuel flow into quantity and price is an example of a hybrid computer.

Characteristics:

- Can handle both **analog** and **digital data**.
- **Real-time processing** with high accuracy.
- Often used in applications where both types of data are needed.

Examples:

- **Petrol pump processors** that convert fuel flow measurements into digital format (quantity and price).
- **Medical equipment** like ECG (Electrocardiogram) machines

4) Tablet and Smartphones

Tablets and Smartphones are the types of computers that are pocket friendly and easy to carry is these are handy. This is one of the best uses of modern technology. These devices have better hardware capabilities, extensive operating systems, and better multimedia functionality. smartphones and tablets contain a number of sensors and are also able to provide wireless communication protocols.

Characteristics:

- **Portable** and **compact** in size.
- **Touch-screen** interface and multiple sensors.
- **Wireless communication** protocols (Wi-Fi, Bluetooth, etc.).
- Capable of running a wide variety of apps for **productivity**, **entertainment**, and **communication**.

Examples:

- **Smartphones** – iPhone, Samsung Galaxy
- **Tablets** – iPad, Samsung Galaxy tablet.

COMPUTER LANGUAGE

You know that software is a set of computer programs. These computer programs will be written in the language what computer understands. If you want to get something done by a person, you will tell him what to do in a language that he understands. Similarly, if you want to get some work done by the computer, you have to tell the computer in a language that the computer understands, i.e., machine language. The machine language consists of only binary digits, i.e. 0 and 1. It was felt quite difficult and tedious for human beings to think in binary numbers. For communicating with the computer, it was thought that it is advisable to develop a third language, a programming language that can be understood by both human beings and the computer. Thus, a programming language is a set of rules that provides a way of instructing the computer to perform certain operations.

Programming languages are said to be lower or higher, depending on whether they are closer to the language the computer itself uses (lower, which means 0s and 1s) or to the language that people use (higher, which means more English like).

The languages in which programs are written are called programming languages. These languages can be classified into following categories.

- Machine language
- Assembly language
- High level language

Machine Languages

Do you know that computer can understand 0's and 1's? Yes 0's and 1's can be directly understood by computers. Programs that have only binary digits are called a machine language programs. It is difficult to write or understand machine language. Letters of the alphabet are also represented in binary numbers. In one system, the letter A is represented as 1000001. Commas, semicolons and other special characters are also represented as bunches of 0s and 1s.

Assembly Languages

In the 1950s, to reduce programming complexity and provide some standardization, assembly languages were developed. Assembly languages, also known as symbolic languages use abbreviations or mnemonic code - codes that are more easily memorized to replace the 0s and 1s of machine languages.

The hypothetical machine language segment we saw above is as follows:

```
11110010 01110011 11010010 00010000 01110000 00101011
```

This could be expressed in assembly language statement as :

```
PACK 210 (8, 13), 02B (4,7)
```

Actually, assembly languages do not replace machine languages. In fact, for an assembly language program to be executed, it must be converted to machine code. The assembly language program is referred to as a 'source program' whereas, the machine language program is an 'object program'.

Assembly language code is very similar in the form to machine language code. In fact, assembly languages had a one-to-one correspondence which means that 15 assembly statements, would be translated into 15 machine language statements. This one-to-one correspondence was still laborious. However, assembly language instructions (called macro instructions) were devised, which executed batches of one-to-one instructions.

Assembly languages offer several advantages:

- They are more standardized and easier to use than machine languages.
- They operate very efficiently, although not as efficient as the machine languages.
- They are easier to debug.

However, there are still some disadvantages:

- Assembly language programs are usually very long and difficult to debug.
- Though less abstract than machine languages, assembly language programs are still complex.

- Though more standardized than machine languages, assembly languages are still machine dependent.

High Level Languages

High Level Languages helped programmers by reducing further the number of computer operations details they had to specify, so that they could concentrate more on the logic needed to solve the problem.

Basic, C, C++, Java, Perl and COBOL are the example of high-level language (HLLL)

Languages are often referred to as generations, the idea being that machine languages were the first generation and assembly languages were the second generation. High-level languages are sometimes used to refer all languages above the assembly level. Here we will subdivide high-level languages into three generation.

- Procedural-oriented or third generation
- Problem-oriented or fourth generation
- Natural or fifth generation

Procedural-oriented Languages

High-level languages are often classified according to whether they solve general problems or specific problems. General-purpose programming languages are called procedural languages or third generation languages. These are languages such as Pascal, BASIC, COBOL, and FORTRAN, which are designed to express the logic, the procedure, of a problem. Because of their flexibility, procedural languages are able to solve a variety of problems.

Procedural languages have many advantages over machine and assembly languages:

- The program statements resemble English language and hence are easier to work with.
- Because of their English-like nature, less time is required to develop a program for a given problem.
- Once coded, programs are easier to understand and modify.
- The procedural languages are machine-independent.

However, procedure-oriented languages still have some disadvantages compared to machine and assembly languages:

- Programs are executed more slowly.
- The languages use computer resources less efficiently.

Problem-oriented Languages and Application Generators

Third-generation languages, such as BASIC or Pascal, require you to instruct the computer in step-by-step fashion. Fourth-generation languages, also known as problem-oriented languages, are high-level languages designed to solve specific problems or develop specific applications by enabling you to describe what you want rather than step-by-step procedures for getting there.

Table 1: summarizes some of the major difference between third-generation languages (3GLs) and fourth-generation languages (4GLs).

Table 1: Difference between 3 GLs and 4 GLs

<i>Third-generation Language</i>	<i>Fourth-generation Language</i>
• <i>Intend for use by professionals</i> • <i>Requires specification of how to perform task.</i> • <i>All alternatives must be specified.</i> • <i>Require large number of procedural instructions.</i> • <i>Code may be difficult to read, understand and maintain.</i> • <i>Language developed for batch operations.</i> • <i>Can be difficult to learn.</i> • <i>Difficult to debug.</i> • <i>Typically file-oriented.</i>	• <i>May be used by non-programming personnel as well as a professional programmer.</i> • <i>Requires specification of what task is to be performed (system determines how to perform the task).</i> • <i>Default alternatives are built in; an end user need not specify these alternatives</i> • <i>Require fewer instructions.</i> • <i>Code is easy to understand and maintain because of English-like commands.</i> • <i>Language developed primarily for online use.</i> • <i>Easy to learn.</i> • <i>Easy to debug.</i> • <i>Typically, database-oriented.</i>

Natural Languages

Natural languages are still in the developmental stages, but they promise to have profound effect, particularly in the areas of artificial intelligence and expert systems. Natural languages have two characteristics:

- They are designed to make the connections that humans have with computers more natural - more human like.
- They are designed to allow the computer to become “smarter” - to actually simulate the learning process by remembering and improving upon earlier information.

Two popular natural languages are LISP and PROLOG.

Binary Logic

In the previous lesson, you have learnt about computer fundamentals. In this lesson you will learn about binary logic. In digital computers, everything (including letters, words and whole texts) is represented in binary numbers (zero and one) i.e., it stores data in terms of bits. Binary logic is the basis of electronic systems, such as computers and cell phones. It works on 0's and 1's. It involves addition, subtraction, multiplication, division of zeros and ones. It includes logic gate functions, AND, OR and NOT which translates input signals into specific output. It facilitates computing, robotics and other electronic applications. In this lesson you will learn about binary numbers, octal numbers, hexadecimal numbers, conversion from one number system to another, character coding system and logic gates.

Number System

You know that computer stores and processes two basic types of data viz., character and number. The character type data includes alphabets and some special symbols, while the number data type includes numerical data on which arithmetic calculations can be done. For example, student name contains character data, roll number will have numeric data.

Digital computers internally use the binary (base 2) number system to represent data and perform arithmetic calculations. The binary number system is very efficient for computers, but not for humans. Representing even relatively small numbers with the binary system requires working with long strings of ones and zeroes.

Binary numbers

Binary numbers are combination of two basic numbers (zero and one) while decimal numbers are combinations of 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9. All the numbers which are generated by decimal numbers can also be generated by Binary numbers. Binary number has base 2 because all numbers are generated by combination of only two numbers (0 and 1) and decimal number has base 10 because all numbers are generated by combination of only ten numbers (0 to 9).

Octal Number and Hexadecimal numbers

Octal (base 8) was previously a popular choice for representing digital circuit numbers in a form that is more compact than binary. Octal is sometimes abbreviated as oct.

Octal counting starts at 0 and goes: 1, 2, 3, 4, 5, 6, 7, 10, 11, 12, 13 14 15 and so on.

Hexadecimal (base 16) is currently the most popular choice for representing digital circuit numbers in a form that is more compact than binary. Hexadecimal numbers are sometimes represented by preceding the value with '0x', as in 0x1B84. Hexadecimal is sometimes abbreviated as hex.

Hexadecimal counting goes: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F, 10, 11, and so on.

The following table shows the decimal number 0 to 15 and its corresponding binary, octal and hexadecimal values.

Decimal, Binary, Octal and Hexadecimal values

Decimal	Binary	Octal	Hexadecimal
0	0000	000	0
1	0001	001	1
2	0010	002	2
3	0011	003	3
4	0100	004	4
5	0101	005	5
6	0110	006	6
7	0111	007	7
8	1000	010	8
9	1001	011	9
10	1010	012	A
11	1011	013	B
12	1100	014	C
13	1101	015	D
14	1110	016	E
15	1111	017	F

All four number systems are equally capable of representing any number. Furthermore, a number can be perfectly converted between the various number systems without any loss of numeric value.

Conversion from one system to another

1. Decimal to Binary Conversion

There are two methods you can use: successive division and subtracting values using a table. Successive division requires dividing continuously by the base we are converting to until the quotient equals 0. The remainders compose the answer.

Steps for conversion from Decimal to Binary

1. Divide the decimal number by 2.
2. Take the remainder and record it on the side.
3. Repeat until the decimal number cannot be divided into any more.
4. With the bits, record them in order from right to left as that will be the number in base 2.

For example, convert 9 into binary as follows;

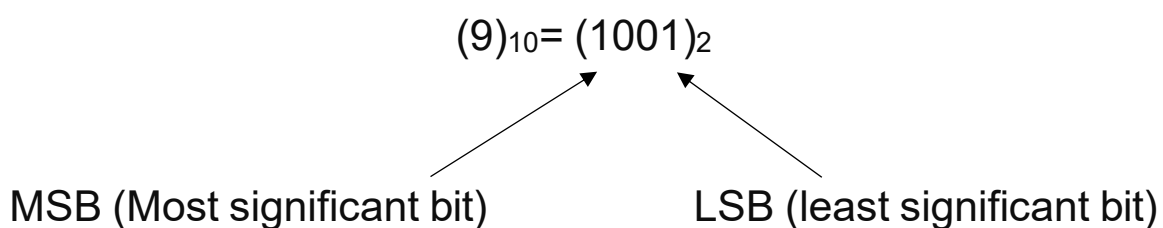
$9/2 = 4$ and remainder = 1 first remainder is called LSB (least significant bit)

$4/2 = 2$ and remainder = 0

$2/2 = 1$ and remainder = 0

$1/2 = 0$ and remainder = 1 last remainder is called MSB (Most significant bit)

Now, decimal 9 is equivalent to 1001 binary number



2. Decimal to Octal Conversion

We can convert decimal to octal and hexadecimal by using the similar conversion method of decimal to binary. i.e., successive division, which requires dividing continuously by the base we are converting to until the quotient equals 0. The remainders compose the answer.

Steps for conversion from Decimal to octal

1. Divide the decimal number by 8.
2. Take the remainder and record it on the side.
3. Repeat until the decimal number cannot be divided any more.
4. With the bits, record them in order from right to left as that will be the number in base 8.

For example, to convert 19 into Octal follow the steps:

$19/8=2$ and remainder = 3 first remainder is called LSB (least significant bit)

$2/8=0$ and remainder = 2 last remainder is called MSB (Most significant bit)

Now decimal number 19 is equivalent to octal number 23

$$(19)_{10} = (23)_8$$

3. Decimal to Hexadecimal Conversion

Division method can be used for conversion from decimal to hexadecimal.

Steps for conversion from Decimal to Hexadecimal

1. Divide the decimal number by 16
2. Take the remainder and record it on the side.
3. Repeat until the decimal number cannot be divided into any more.
4. With the bits, record them in order from right to left as that will be the number in base 16.

For example, to convert decimal number 229 into hexadecimal number:

$229/16 = 14$ and remainder = 5 first remainder is called LSB (least significant bit) $14/16 = 0$ and remainder = 14 last remainder is called MSB (Most significant bit)

In hexadecimal 10,11,12,13,14,15 are equivalent to A,B,C,D,E,F respectively

So $(229)_{10} = (E5)_{16}$

4. Binary to decimal

To find the decimal representation of a binary number simply take the sum of products of binary digits and the powers of 2 which they represent.

For example,

Conversion of $(1000)_2$ into decimal is as follows:

$$(1000)_2 = 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$

$$= 1 \times 8 + 0 \times 4 + 0 \times 2 + 0 \times 1$$

$$(1000)_2 = 8 + 0 + 0 + 0 = 8$$

5. Octal to decimal

To find the decimal representation of an octal number simply take the sum of products of octal digits and the powers of 8 which they represent.

For example,

Conversion of $(1000)_8$ into decimal is as follows:

$$(1000)_8 = 1 \times 8^3 + 0 \times 8^2 + 0 \times 8^1 + 0 \times 8^0$$

$$= 1 \times 512 + 0 \times 64 + 0 \times 8 + 0 \times 1$$

$$(1000)_8 = 512 + 0 + 0 + 0 = 512$$

6. Hexadecimal to decimal

To find the decimal representation of a Hexadecimal number simply take the sum of products of hex number and the powers of 16 which they represent.

For example,

Conversion of $(1000)_{16}$ into decimal is as follows:

$$(1000)_{16} = 1 \times 16^3 + 0 \times 16^2 + 0 \times 16^1 + 0 \times 16^0$$

$$= 1 \times 4096 + 0 \times 256 + 0 \times 16 + 0 \times 1$$

$$(1000)_{16} = 4096 + 0 + 0 + 0 = 4096$$

Now let us learn about one's complement and two's complement of a binary number.

ONE'S COMPLEMENT AND TWO'S COMPLEMENT

One's complement of a binary number can be achieved by changing all 0's to 1's and 1's to 0's.

Example:

There is a binary number 11001 and its 1's complement would be 00110 after all

0's to 1's and 1's to 0's.

Two's complement of binary number can be obtained by adding 1 to the LSB (Least Significant Bit) of 1's complement of binary number.

2's complement = 1's complement + 1

Example:

Binary number: 11001

1's complement of number: 00110 (i.e. convert all 1's to 0's and all 0's to 1's)

2's complement of number = $00110 + 1 = 00111$

CHARACTER CODING SYSTEM (ASCII, ISCII & UNICODE)

Computer works on character data and this data is not only alphabet but numeric values, punctuation; spaces, etc., are also character data. In common language whatever is on keyboard (except shift, caps lock key) are character data. As you know computer operates on binary values so these characters are also represented in binary values. Most common character coding used in India are ASCII, ISCII and UNICODE.

ASCII

ASCII stands for American Standard Code for Information Interchange.

Computers can only understand numbers, so an ASCII code is the numerical representation of a character such as 'A' has numerical value as 65. Some other characters and its equivalent ASCII values are following:

Table 1.1: ASCII codes table

ASCII	Hex	Symbol
64	40	@
65	41	A
66	42	B
67	43	C
68	44	D
69	45	E
70	46	F
71	47	G
72	48	H
73	49	I
74	4A	J
75	4B	K
76	4C	L
77	4D	M
78	4E	N
79	4F	O

From the above table you can see, characters have equivalent ASCII values. Similarly, each character has ASCII value.

ASCII value for a to z is
from 97 to 122 i.e., a = 97, z = 122.

Indian Script Code for Information Interchange (ISCII)

This coding scheme is used for Indian script and its symbols. Using this, it is possible to define a syllable boundary for Indian script word. It was developed by Bureau of Indian Standards in 1986 and revised to a more compact form in 1988. The table 1.2 shows few of the character set for Devanagari script -

Table 1.2: “first vowel of Hindi has numeric value 164”

अ 0905 164	आ 0906 165	इ 0907 166	ई 0908 167
उ 0909 168	ऊ 090A 169	ऋ 090B 170	ऐ 090E 171
क 0915 179	ख 0916 180	ग 0917 181	घ 0918 182
० 0966 241	१ 0967 242	२ 0968 243	३ 0969 244

LOGIC GATES

Logic gates are elementary building block of a digital circuit. Most logic gates have two inputs and only one output. The basic logic gates are AND, OR and NOT, NAND, NOR, XOR and XNOR. In this lesson, let us learn about AND, OR, NOT, NAND and NOR logic gates. Binary logic deals with true and false. You know that 0 is usually associated with false and 1 is associated with true.

Example

- (a) $1 + 0 = 1$ (True)
- (b) $0 + 1 = 1$ (True)
- (c) $0 + 0 = 0$ (False)
- (d) $1 + 1 = 1$ (False)

1.AND gate

This gate acts in the same way as the logical operator 'and' works. An AND operation produces an output of logic-0 if at least one of the inputs is zero. Output will be logic-1 if all the inputs are logic-1. Truth table of AND operation is shown below. A truth table is a list of all possible combination of input logic levels along with corresponding output level (refer to Table 1.1).

Table 1.1: Truth Table for AND operation

Input		Output
A	B	C=A.B
0	0	0
0	1	0
1	0	0
1	1	1

The equation for logic AND operation

$$C = AB$$

AND gate representation

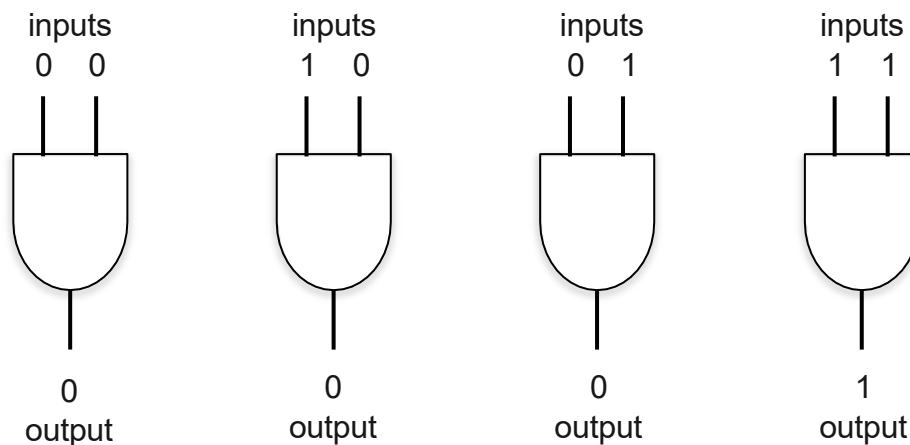


Fig. 1.1: AND gate

2. OR gate

This operation is represented by plus (+) sign. The OR gate produces an output of logic-1 if at least one of the inputs is logic-1.

Truth table of OR operation is given in Table 2.1

Table 2.1: Truth Table for OR operation

Input		Output
A	B	$C = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

Equation of OR operation: $C = A + B$

OR gate representation

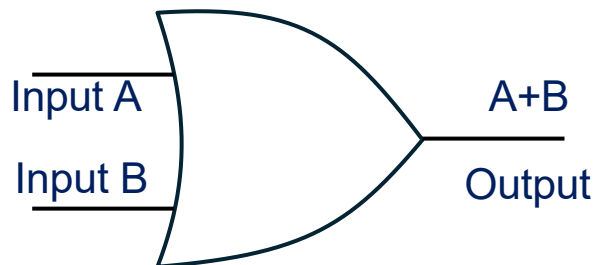


Fig. 2.1: OR Gate representation

In the above representation A,B are inputs and the output is A+B.

3.NOT operation

The NOT operation represents complement or inverse of inputs. A prime or bar represents this operation. The truth table for NOT operation is given in Table 3.1.

Table 3.1: Truth table for NOT operation

(Input)	(Output)
A	$\bar{A}$
0	1
1	0

NOT gate representation

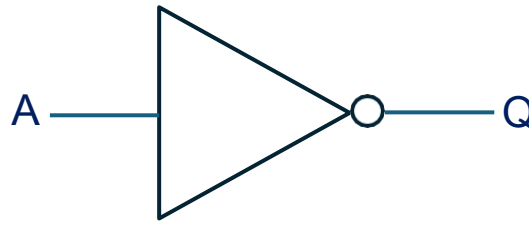


Fig. 3.1: NOT Gate representation

Equation for NOT operation

$$Q = \bar{A}$$

Other operations are combination of AND, OR and NOT. They are NOR and NAND.

- NOR operation is combination of OR and NOT. NAND operation is combination of AND and NOT.
- NAND and NOR operations are also known as universal operations because

you can generate any other operation through it. These gates are called Universal Gates.

4.NOR Operation

It is the NOT-OR gate. The output is true when neither A nor B is true. If any one of the input variables is true then output is false. NOR gate represents complement of the OR operation. The graphic symbol for the NOR gate consists of an OR symbol with a bubble on the output, denoting that a complement operation is performed on the output of the OR gate.

Table 4.1: Truth table of NOR operation

Input		Output
A	B	$C = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

NOR gate representation

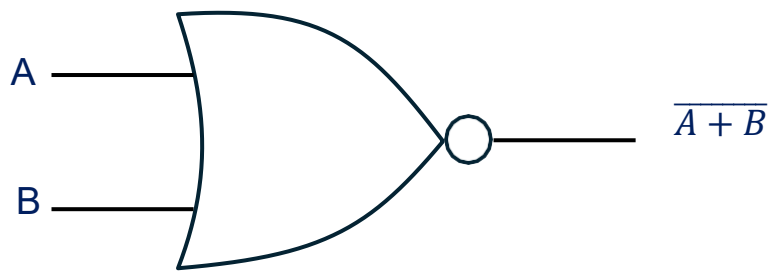


Fig. 4.1: NOR Gate representation

Equation for NOR gate

$$C = (A + B)^1 \text{ or } \overline{A + B}$$

5.NAND Operation

The NAND gate represents the complement of the AND operation. The graphic symbol for the NAND gate consists of an AND symbol with a bubble on the output, denoting that a complement operation is performed on the output of the AND gate. Truth table of NAND operation is given in Table 5.1.

Table 5.1: Truth table of NAND operation

Input		Output
A	B	$C = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

NAND gate representation

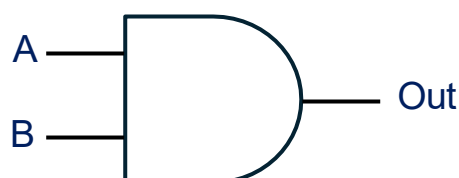


Fig. 5.1: NAND Gate representation

Equation for NAND gate

$$C = \overline{A \cdot B} \text{ or } (A \cdot B)'$$

NAND and NOR gates are called universal gates as they can implement any type of boolean expression.

Common Computer Shortcut Keys

Shortcut Key	Description
Ctrl + C	Copy selected text or item
Ctrl + X	Cut selected text or item
Ctrl + V	Paste copied or cut content
Ctrl + Z	Undo the last action
Ctrl + Y	Redo the last undone action
Ctrl + A	Select all text or items
Ctrl + S	Save the current document
Ctrl + P	Print the current document
Ctrl + F	Open the Find dialog box
Ctrl + H	Open Find and Replace dialog box
Ctrl + B	Make the selected text bold
Ctrl + I	Make the selected text <i>italic</i>
Ctrl + U	Underline the selected text
Alt + Tab	Switch between open applications
Alt + F4	Close the active application or window
Win + D	Show or hide the desktop
Win + L	Lock the computer screen

Full Forms

(Computer Fundamentals)



Memory & Storage

RAM = Random Access Memory

ROM = Read Only Memory

HDD = Hard Disk Drive

SSD = Solid State Drive

CD = Compact Disc

DVD = Digital Versatile Disc

BD = Blu-ray Disc



Hardware & System

CPU = Central Processing Unit

ALU = Arithmetic Logic Unit

CU = Control Unit

BIOS = Basic Input Output System

SMPS = Switched Mode Power Supply

UPS = Uninterruptible Power Supply

GPU = Graphics Processing Unit



Input & Output Devices

VDU = Visual Display Unit

CRT = Cathode Ray Tube

LCD = Liquid Crystal Display

LED = Light Emitting Diode

HDMI = High Definition Multimedia Interface

VGA = Video Graphics Array



Networking

LAN = Local Area Network

MAN = Metropolitan Area Network

WAN = Wide Area Network

PAN = Personal Area Network

IP = Internet Protocol | **MAC**

MAC = Media Access Control



Software & File Formats

OS = Operating System

GUI = Graphical User Interface

CLI = Command Line Interface

PDF = Portable Document Format

DOC = Document

XLS = Excel Spreadsheet



Image & Multimedia Formats

JPEG / JPG = Joint Photographic Experts Group

PNG = Portable Network Graphics

GIF = Graphics Interchange Format

BMP = Bitmap

MP3 = MPEG Audio Layer-3

PPT = PowerPoint Presentation

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